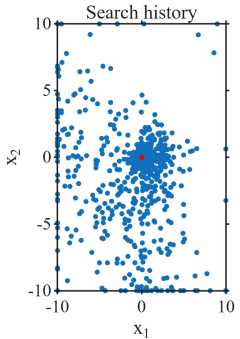
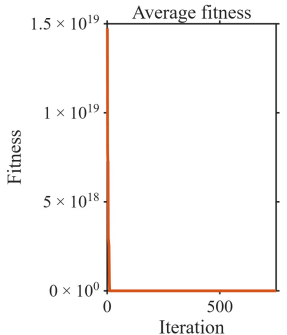
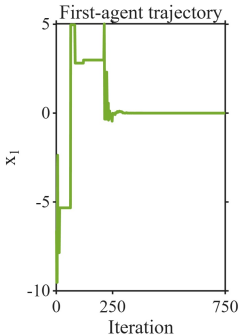
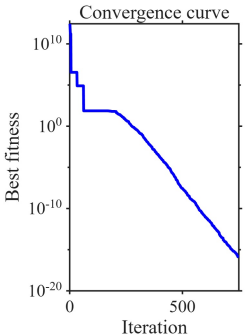

F2
Fitness
10000
5000
0
100
0
-100
x2
-100
0
100
x1


Search history
x2
10
5
0
-5
-10
-10
0
10
x1


Average fitness
Fitness
1.5 × 10^19
1 × 10^19
5 × 10^18
0 × 10^0
0
500
Iteration


First-agent trajectory
x1
5
0
-5
-10
0
250
750
Iteration


Convergence curve
Best fitness
10^10
10^0
10^-10
10^-20
0
500
Iteration